

Practice Questions: Numerical Integration using SciPy

Topic: Single and Double Integration

Objective: Learn how to compute definite integrals using SciPy using `quad()` and `dblquad()`.

Question 1: Single Integration of a Polynomial Function

Evaluate: $\int_0^2 x^2 + 5x + 6 \, dx$

- Step 1: Import `quad` from `scipy.integrate`.
- Step 2: Define the function $f(x) = x^2 + 5x + 6$
- Step 3: Use `quad()` from 0 to 2.
- Step 4: Store the result and estimated error.
- Step 5: Display both values.

Question 2: Single Integration of a Trigonometric Function

Evaluate: $\int_0^{\pi/2} \sin^2(x) \, dx$

- Step 1: Import NumPy and `quad`.
- Step 2: Use NumPy's `sin()` function.
- Step 3: Apply `quad()` between 0 and $\pi/2$
- Step 4: Print the result and error estimate.

Question 3: Single Integration using a Lambda Function

Evaluate: $\int_1^3 (x^3 + 2x) \, dx$

- Step 1: Import `quad`.
- Step 2: Create the function using a lambda expression.
- Step 3: Use `quad()` with limits 1 and 3.
- Step 4: Display the integral value.

Question 4: Single Integration of an Logarithmic Function

Evaluate: $\int_0^4 \ln(x) \, dx$

- Step 1: Import NumPy and `quad`.
- Step 2: Define the log function.
- Step 3: Use `quad()` to compute the integral.
- Step 4: Print the result and estimated error.
- Step 5: Use `dblquad()` and display result and error.

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Question 5: Double Integration of a Polynomial Function

Evaluate: $\int_0^2 \int_0^1 (x^2 + y^2) dy dx$

- Step 1: Import dblquad.
- Step 2: Define the function $f(y, x)=x^2+y^2$.
- Step 3: Set x limits from 0 to 2.
- Step 4: Set y limits from 0 to 1.
- Step 5: Perform the integration and print the value.

Question 6: Double Integration for Area Calculation

Evaluate: $\int_0^3 \int_0^2 (x + y) dy dx$

- Step 1: Import dblquad.
- Step 2: Define the function $f(y, x)=x+y$.
- Step 3: Set x limits from 0 to 3.
- Step 4: Set y limits from 0 to 2.
- Step 5: Use dblquad() and display result and error estimate.

Practice Requirements

1. Write the complete Python program.
2. Add comments explaining each step.
3. Display the result clearly.
4. Display the estimated error returned by SciPy.
5. Verify the answer manually wherever possible.